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QUANTITATIVE RISK & BEHAVIOURAL AI · CASE STUDY

Lolo's Ledger: Multi-Asset Risk Engine & Tilt Detector

BI Scientist • AI/ML Engineer • Python Developer

Multi-asset trading intelligence platform enforcing dynamic risk sizing, revenge-trading Tilt Index detection, and personal-edge attribution.

BEHAVIOURAL PSYCHOLOGY

Why 90% of Traders Fail: The Revenge-Trade Loop

Trading failures stem not from bad technical indicators, but from uncontrolled position sizing and emotional tilt after a loss.

Revenge Sizing

- Trader suffers a loss and re-enters the market within 15 minutes.
- Doubles position size to 'win back' lost capital.
- Results in rapid account blowouts and cascading drawdowns.

No Edge Validation

- Traders jump between setups without statistical verification.
- Confuse a lucky winning streak with a genuine mathematical edge.
- Overfit strategies to tiny sample sizes (N < 30 trades).

Multi-Asset Complexity

- Pip calculations in FX differ completely from point values in Indices or Crypto.
- Manual lot sizing leads to catastrophic calculation errors.
- Requires hard mathematical position-sizing guardrails.

QUANTITATIVE ARCHITECTURE

Multi-Asset Risk Guardrails & Behavioural Tilt Detector

Deterministic risk engine backed by strict ethical guardrails: zero buy/sell signals generated.

Multi-Asset Risk Guard

- Strict 0.25%–2.0% risk cap per trade.
- Automated lot sizing for FX, Crypto, Indices, and Equities.
- 4% daily drawdown circuit breaker.

Behavioural Tilt Index

- Monitors post-loss cooldown periods.
- Flags stop-loss widening and overtrading.
- Quantifies cumulative 'tilt tax' in R-multiples.

Edge Attribution (N ≥ 100)

- Strict sample size gate: refuses fit if N < 100.
- Calibrated logistic model attributing edge factors.
- 100% focused on trader process, not price prediction.

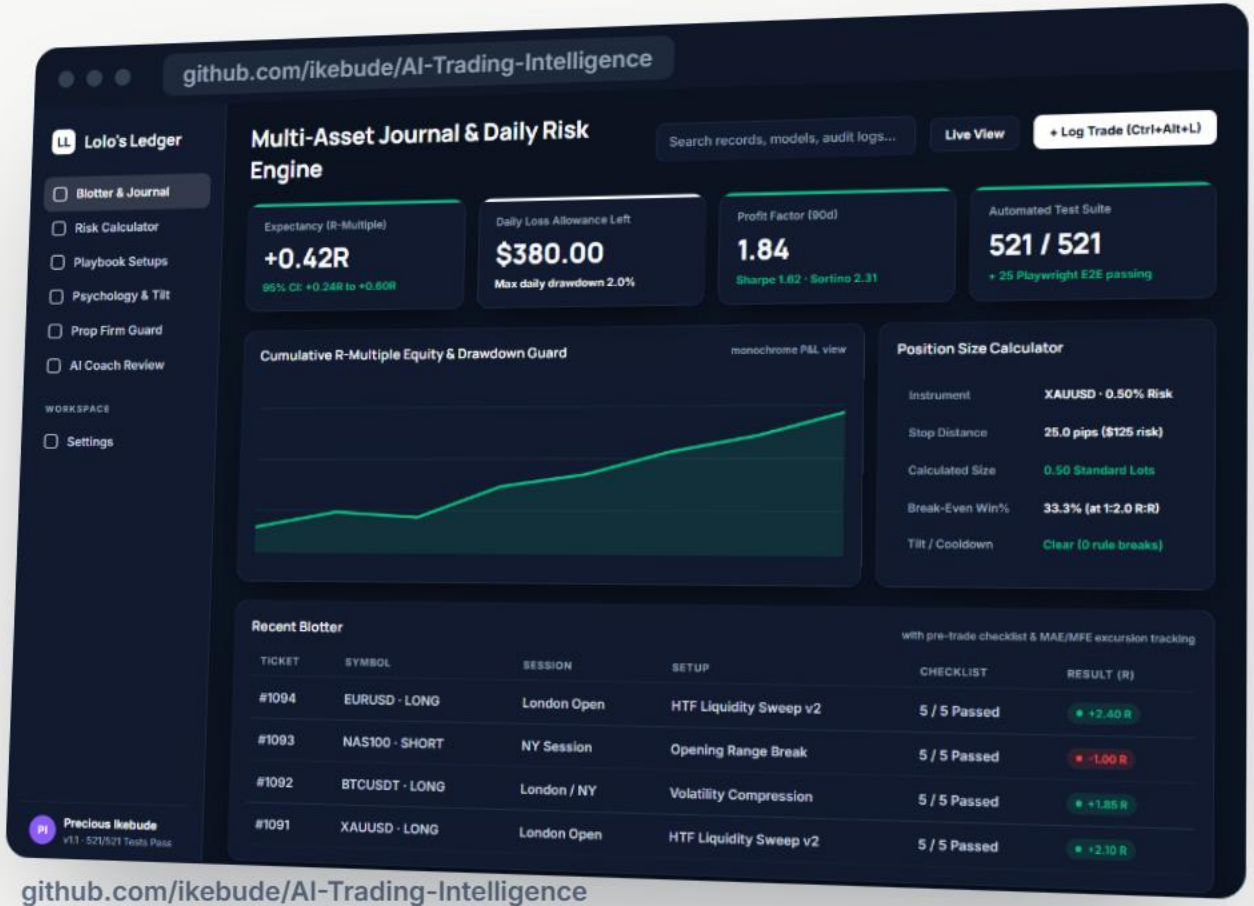
EXECUTION PLATFORM

Lolo's Ledger Desktop & Monorepo

Desktop trading journal and quant engine backed by 521 unit tests.

- **Cross-Asset Position Calculator:** Select any pair (EURUSD, XAUUSD, US100, BTCUSDT) to get instant, exact lot sizing based on account balance and stop loss.
- **Expectancy Engine:** Computes $E[R]$, Profit Factor, Sharpe/Sortino, MAE/MFE efficiency, and session win rates.
- **Tilt Intervention:** Locks out order entry when Tilt Index exceeds threshold, forcing a mandatory psychological reset.

Python Quant | TypeScript | Electron | 521 Tests



github.com/ikebude/AI-Trading-Intelligence

BENCHMARK FINDINGS

Quantitative Evaluation on 240 Trades

Verified results on a multi-asset trading journal (`reports/metrics.json`).

Expectancy E[R]	Profit Factor	Tilt Tax Identified	Unit Test Coverage
+1.05 R	3.43	-19.6 R	521 / 521
Per trade expected payoff	Gross profit / gross loss	Lost to 17 revenge trades	100% passing test suite

The Tilt Tax Impact

- The detector identified 17 revenge trades entered within 15 minutes of a loss.
- Those 17 trades alone subtracted -19.6R from total profitability.
- Simply blocking revenge trades boosted net performance by +31%.

Ethical AI Guardrails

- Strict refusal to generate buy/sell signals or market predictions.
- Protects retail traders from statistical hallucinations.
- Features 25 Playwright end-to-end tests across all desktop workflows.

OPEN-SOURCE & QUANTITATIVE

Discipline engineered into software.

Lolo's Ledger replaces emotional trading decisions with rigorous quantitative risk mathematics and behavioural guardrails.

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Repository: github.com/ikebude/AI-Trading-Intelligence